



Development of a chemical selective iron Fischer Tropsch catalyst



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ABSTRACT

In the chemical industry C₂–C₄ olefins are critical building blocks for high value commodity products. One of the key challenges in Fischer Tropsch Synthesis (FTS) technology development is to manipulate process conditions and/or catalytic systems in order to suppress methane selectivity while still producing mainly C₂–C₄ products with maximum olefin content. Sasol developed an iron catalyst promoted with sodium and sulphur that yielded a fairly high proportion of C₂–C₄ products with lower than expected methane selectivity. Due to the unexpected low methane selectivity in combination with high olefin selectivity necessitated the evaluation of other promoters similar to sulphur in combination with alkali promoters (*i.e.* sodium and potassium) in order to understand whether the combination of sulphur and sodium is unique or whether other alkali and non-metal promoters will yield similar results or even improve the selectivity further. By employing high throughput experimentation, a total of 306 catalysts were synthesised and tested in order to identify whether other possible non-metal FTS promoters could either match or improve the selectivities observed for sodium and sulphur and to also identify potential optimum promoter levels for maximum C₂–C₄ olefin selectivity and minimum methane formation. The results showed that from the series of catalysts evaluated sulphur and sodium yielded the highest olefin selectivity in combination with low methane selectivity and that a mole ratio of at least 2:1 Na:S is required for optimum C₂–C₄ selectivity. Apart from the statistical analysis of data and subsequent comparison of catalyst selectivity, the catalysts were also evaluated based on the expected Schulz Flory selectivity correlation. For unique catalyst selectivity a significant shift should be observed. In this paper the results of high throughput experimentation are presented and discussed using the direct relations of C₂–C₄ selectivity and methane selectivity in order to distinguish unique catalyst systems that are selective towards C₂–C₄ olefin production.

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1. Introduction

Iron-catalysed Fischer Tropsch Synthesis (FTS) has been used commercially from the early 1950s by Sasol in South Africa to produce fuel and commodity chemicals such as wax. The advantage of this technology is that the product spectrum can be manipulated either *via* process conditions or choice of catalyst system to yield a range of high value products *e.g.* waxes, liquid hydrocarbon fuels and chemicals *e.g.* olefins and oxygenates. The wide range of products produced by FTS can be advantageous especially if product types and volumes could be manipulated to capitalise on current market demand. Due to recent fluctuations in the oil price the need for flexibility in product selectivity is becoming more critical and could provide opportunities to shift production to higher value

commodities. Ideally, this should be achieved using existing hardware with limited modifications and capital expenditure. Research should thus focus on possible manipulation of the catalyst system and/or process conditions. Sasol uses both low and high temperature iron-based FTS in its operations, with the high temperature process focused on producing liquid fuel and commodity chemicals while the low temperature process produces mainly waxes. Although Sasol has extensive knowledge of maximising chemical production from iron catalysed Fischer Tropsch Synthesis, the main challenge remains producing commodity chemicals such as C₂–C₄ olefins with minimum methane and paraffin selectivity. In the chemical industry C₂–C₄ olefins are critical building blocks for high commodity products and one of the key challenges in FTS technology development is to manipulate process conditions and/or catalytic systems in order to suppress methane selectivity while maximising C₂–C₄ olefin products. The ideal catalyst should also exhibit high activity and allow for high CO conversion rates. The Schulz Flory distribution is a mathematical function that is

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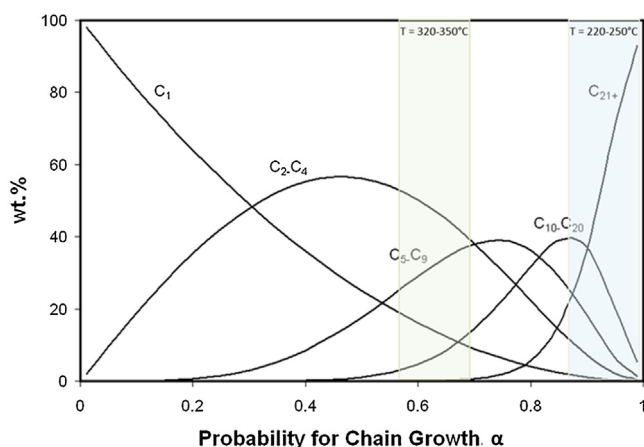


Fig. 1. Influence of α -value on product distribution. Highlighted areas show the influence of temperature on the selectivity of iron based Fischer Tropsch Synthesis.

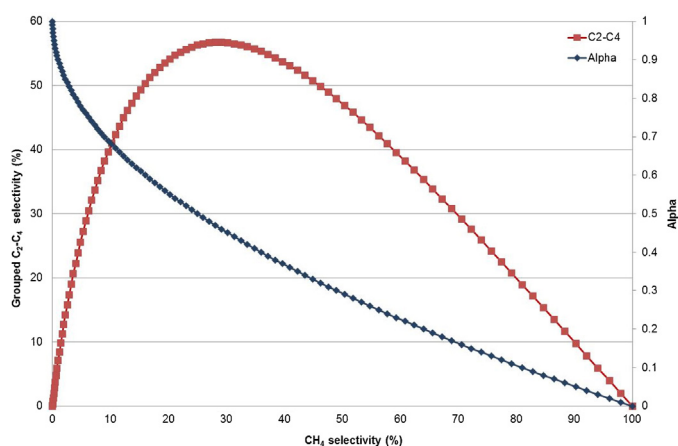


Fig. 2. Fixed relationship between grouped C₂ to C₄ selectivity and methane selectivity as predicted by the ideal Schulz-Flory model.

generally used to describe and/or predict the selectivity profile for various iron FT catalysts. It describes the probability of chain growth on specific catalyst sites. Fig. 1 shows the selectivity of various product fractions as a function of chain growth probability. For low α FT catalysts ($\alpha < 0.5$) it is shown that the C₂–C₄ selectivity increases, through a maximum C₂–C₄ weight percent with a concomitant decrease in methane selectivity. As the α value increases beyond 0.5 both methane and C₂–C₄ selectivity are inhibited. The product selectivity as described by the Schulz Flory correlation is also responsive to variation of process parameters *i.e.* temperature, pressure and H₂/CO ratio and predicts product profiles for most known FTS catalysts. For lower temperature iron FTS catalysts α values as high as 0.94 is observed yielding mainly wax products.

Based on the Schulz Flory distribution the challenge is to identify a catalytic surface which will both increase hydrogenation rate *i.e.* limit chain growth while limiting paraffin and specifically methane selectivity. From Fig. 1 it can be observed that the higher the temperature, the lower the chain growth probability of the catalyst, but this has a significant impact on the effectiveness of the FTS process *i.e.* high volumes of methane being produced that needs to be recycled back into the system. The current high temperature iron FTS process produces only ~40% of C₂–C₄ products (mainly paraffins) with methane selectivities of ~10–15%. The ideal Schulz Flory correlation of iron FTS catalysts predicts a fixed relationship between methane and the grouped C₂–C₄ selectivity

(Fig. 2). The corresponding α values are also presented in this graph. As the probability of chain growth decreases, the methane selectivity increases up to 100% for a catalyst with an α value of zero. The grouped C₂–C₄ selectivity goes through a maximum at a corresponding C₁ selectivity of just below 30%. According to this ideal relationship, at a maximum, iron high temperature FT catalysts can yield a grouped C₂–C₄ selectivity of $\pm 57\%$, but with the cost of an appreciable methane make. Due to the somewhat higher-than-expected methane selectivity usually observed experimentally, the actual grouped C₂–C₄ selectivity is anticipated to lie below the curve of Fig. 2. That is clearly not an ideal situation from an industrial perspective if the aim is to selectively produce light chemicals and optimisation is required in order to alter the fixed relationship between C₂–C₄ olefins and methane from the standard Schulz Flory distribution.

A recent review by Torres and de Jong highlighted the renewed focus on developing a FT catalyst that yields mainly C₂–C₄ olefins [1]. Primarily iron based catalysts have shown the greatest propensity to produce olefins and therefore have been the basis for most studies focused on producing chemicals from syngas. Several studies in literature focused on developing catalyst systems with optimum C₂–C₄ selectivity while minimising methane selectivity [2–4]. These include combinations of metal oxides/sulfides other than iron and cobalt (molybdenum, platinum, nickel, *etc.*) with alkali metals [5]. In the field of iron FTS catalysis it was noted that sulphur in combination with promoters such as manganese, phosphorous, *etc.* could also yield improved C₂–C₄ selectivity [6]. In a study by Van Dijk *et al.* [7] an iron/manganese catalyst containing sulphur showed improved ethylene selectivity at temperatures above 350 °C. However, in spite of the various approaches to obtain an improved yield of short chain products described in literature, the main challenge remains to develop a catalyst that can selectively produce C₂–C₄ olefins, while keeping the methane formation as low as possible. Early work by Bromfield and Coville [8] indicated that the addition of small quantities of sulphide induced a modification of the morphology of the iron oxide catalyst surface resulting in enhanced surface area and porosity and thus yielding higher conversions. In addition to this the sulphide promoted catalysts also yielded a higher percentage of olefins in the product profile. However, the product distribution followed a typical Schulz Flory correlation and did not afford the desired minimum methane selectivity. In this regard Sasol developed an iron FTS catalyst that showed a high propensity towards C₂–C₄ olefins, while simultaneously suppressing excess methane formation, by combining alkali and/or alkaline earth metal promoters with sulphur [9]. The catalyst yielded up to 68% of olefins in the C₂–C₄ fraction with a relatively low methane selectivity of 13% and CO converted to hydrocarbons of 40%. This catalyst showed a high affinity for C₂–C₄ products (~55%) and lower than expected methane selectivity (<15%) and was used as a reference catalyst for further optimisation. Torres Galvis *et al.* [10] confirmed this observed increase in olefin selectivity while evaluating the effect of sodium in combination with sulphur on supported iron FTS catalysts. The group showed that these promoters added to Fe/ α -Al₂O₃ catalysts yielded high C₂–C₄ olefin selectivity (~50%), enhanced catalytic activity, and decreased methane production (<20%) relative to Fe/ α -Al₂O₃ catalysts not promoted with sodium and sulphur.

The question to be answered was whether the combination of sulphur and sodium was unique in achieving these results or whether other possible promoter combinations could yield even higher olefin selectivities. Due to the potential technology impact of an olefin selective FTS catalyst, Sasol decided to expedite the development *via* high throughput experimentation, which is an enabling technology that allows for the rapid and parallel evaluation of numerous catalyst formulations. The hardware associated with high throughput screening has evolved rapidly over recent

years. Compact analysis techniques, automation and increasing miniaturization have led to standalone systems being available to the industry at large. A series of five promoters were selected from the group of non-metals (boron, phosphorous, sulphur, chloride and antimony) to be combined with one of two alkali FT promoters (sodium and potassium) to afford ten groups of catalysts for evaluation. The catalysts were prepared using reverse precipitation of iron nitrate with ammonium hydroxide. The ammonium sulphide was added with the iron nitrate while the alkali carbonate salt was added after precipitation. These catalysts were evaluated and compared to the in-house developed Sasol reference catalyst (referred to in this document as Sasol Reference Na–S catalyst). The absolute promoter levels were then systematically varied through the use of statistical experimental design methods. A total of 306 catalysts were synthesised and tested with the aim of finding the optimum promoter regions *i.e.* those regions yielding maximum C₂–C₄ selectivity, with minimum methane formation, at commercially acceptable conversion levels. Apart from the statistical analysis of data and subsequent comparison of catalyst selectivity, the catalyst was considered to be truly unique if it yielded a product slate that deviated significantly from the ideal Schulz Flory selectivity prediction. In this paper the results of high throughput experimentation are presented and discussed using the direct relations of C₂–C₄ selectivity and methane selectivity (as predicted by the ideal Schulz Flory correlation) in order to distinguish catalyst systems that are selective for C₂–C₄ production.

2. Experimental

2.1. Catalyst preparation

All catalysts were prepared using the method described below. The choice of the non-metal promoters were governed by the following principles: a precursor that would decompose completely after calcination, commercial availability, low toxicity of precursor (avoid poisonous compounds) and solubility of precursor in water (antimony acetate had to be dissolved in citric acid). Table 1 summarises the precursors with concentrations used during this study. All promoter concentrations were calculated based on 100 g Fe (*i.e.* 1.78 mol Fe).

During the preparation procedure Fe(NO₃)₂·9H₂O (1 M) was dissolved in 500 ml of water. The required amount of the non-metal salt solution (see Table 1) was added before precipitation. The pH of the solution was typically ±0.5 and the temperature was ±16 °C. While the mixture was vigorously stirred NH₄OH (25 wt.%) was added to the solution until a pH of ±6 and temperature of ±27 °C was reached. No temperature control was used during the precipitation step. Na₂CO₃ was added as a promoter and the final pH of the solution was around 6.3 and the final temperature was 25 °C. The catalyst was dried in an oven overnight at 150 °C and calcined at 350 °C in a muffle oven for 4 h in air. Finally the catalyst was crushed and sieved to a particle size between 38–150 µm. The amount of non-metal promoter added varied between 0.4 to 5 mmol and the alkali promoter varied between 1.6 and 26 mmol. Chemical

composition of catalysts was determined by atomic adsorption/ICP analysis and selected catalysts were characterised using BET measurements, XRD and Pore Size Distribution.

2.2. Catalyst evaluation

The high-throughput catalyst testing was conducted using the Avantium Nanoflow unit, consisting of 4 phases of 16 fixed bed reactors each. One fixed bed reactor was loaded with inert material while the other fifteen reactors were loaded with 1 ml of calcined catalyst (particle size 38–150 µm, 25 mg) to provide a catalyst bed of ~100 mm. The catalyst was not diluted by an inert medium, but corundum was used to ensure consistent catalyst packing. The feed gas to all reactors was kept the same and due to the configuration of the unit all reactors were heated to the same temperature. The gaseous compounds (CO, CO₂, H₂, light hydrocarbons (C₁–C₈)) were measured online using GC–TCD and GC–FID. The heavier hydrocarbons were collected *via* hot gas condensers and analysed off line by GC–FID and MS analysis. The catalyst was reduced *in situ* at 420 °C for 16 hours at a pressure of 20 bar H₂. After reduction the reactor temperature was set to 330 °C and synthesis gas (composition of H₂ (57 vol%), CO (14 vol%) and CO₂ (11 vol%) balanced with He) was introduced at a flow rate of 12.9 l (n) per g catalyst per hour. The total pressure was kept at 20 bar (see Table 2).

2.3. Experimental design and statistical methodology

Statistical experimental design is a very efficient method of evaluating the effect of many variables on one or more responses, providing the maximum information with the fewest number of experiments. A 23-experiment D-optimal design was used to cover a wide range of promoter levels and combinations (7 levels per combination). The design was selected for estimation of a cubic model in the two design variables. A cubic model was employed *a priori* since it was not clear what type of response would be observed within the experimental ranges tested. Centre points were done in triplicate. Replicating the centre point provides the ability of estimating the experimental error in the program and provides a test for non-linearity in the design range. In the case of 10 promoter combinations (2 alkali and 5 non-metal promoters), a total of 230 catalysts were prepared and tested. The design included 6 replicates: duplicates of the corner points (2 × 4) and a triplicate of the middle point (3 × 1). These points are shown in Fig. 3. Furthermore, it is assumed that the experimental error at all the experimental conditions is the same as the error at the middle point *i.e.* the experiments are performed within a homogeneous environment. This significantly reduces the number of experiments required.

The data were statistically evaluated with the analysis of variance (ANOVA) procedure. The ANOVA simultaneously tests the difference between the means of multiple values relative to their variance for many variables. Furthermore, the ANOVA quantifies which variables have a statistically significant effect on the response of interest. Following the ANOVA analysis, a regression model in the significant variables can be constructed for interpretation of the effect of the variables on the response, and for

Table 1
Summary of precursors used.

Promoter	Precursor	Concentration used
Sulphur	Ammonium sulphate	1.0 M
Boron	Ammonium baborate tetrahydrate	0.3 M
Antimony	Antimony acetate	0.5 M ^a
Chloride	Ammonium chloride	1.0 M
Phosphorous	Ammonium phosphate dibasic	1.0 M
Sodium	Sodium carbonate	Added as powder
Potassium	Potassium carbonate	Added as powder

^a Dissolved in citric acid.

Table 2
Summary of non-metal promoter precursors and concentrations used in the study.

	Condition
Reduction temperature	420 °C
Synthesis temperature	330 °C
Pressure	20 bar
H ₂ :CO ratio	4:1
GHSV	12.9 L(n) g-cat h ⁻¹



Fig. 3. Proposed experimental design for promoter level optimisation.

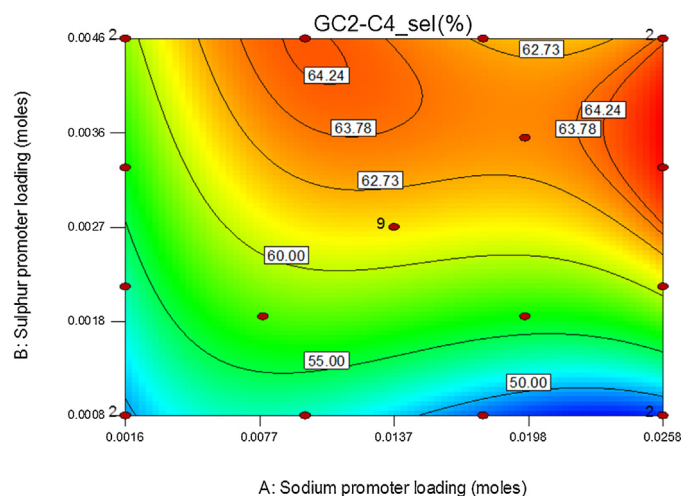


Fig. 4. Contour plot of predicted group C₂–C₄ selectivity (%) as a function of sodium and sulphur promoter loadings.

prediction of the response at any combination of the variables' values in the model. All the design and modelling were done on the Design-Expert® software [11].

3. Results and discussion

3.1. Evaluation of selectivity data

The main objective of this work was to identify optimum promoter concentrations for an iron FTS catalyst to yield maximum lower olefins and minimum methane selectivity by using high throughput experimentation methodology. From a commercial perspective the developed catalyst must be a replacement catalyst for the current high temperature iron FTS process utilised by Sasol with little or no modifications required on the current FT reactor. For this reason the focus was on optimisation of unsupported catalysts that can be commercially produced at relatively low cost. Due

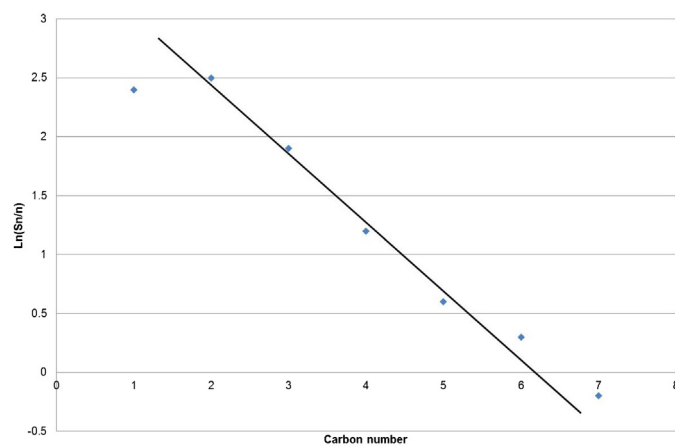


Fig. 5. Schulz Flory plot of a typical product spectrum from the Sasol reference Na–S catalyst.

to the large number of permutations of promoter ratio's and levels possible, high throughput screening aided in optimisation of catalyst formulation within a relatively short period of time. A large number of catalysts with various combinations of sodium and sulphur was prepared and tested. The data was statistically evaluated with the analysis of variance (ANOVA) procedure, which preceded the construction of a regression model of the significant variables. Fig. 4 shows the contour plot of the predicted group C₂–C₄ selectivity (%) as a function of the sodium and sulphur promoter loadings. The maximum C₂–C₄ selectivity observed was 66%.

The statistical design and analysis methodology was successfully applied during the high throughput experimentation in order to optimise not only catalyst composition, but also evaluate optimum process conditions. However, to evaluate whether these catalysts were significantly different compared to traditional iron FTS catalysts it was decided to compare the Schulz Flory carbon distribution of the FTS products. The Sasol reference Na–S catalyst yielded a high selectivity towards light olefins and alcohols without excessive methane formation. A typical

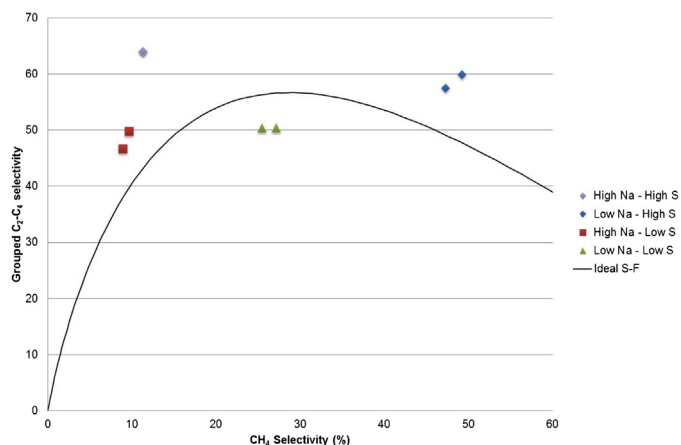


Fig. 6. Results obtained with various combinations of sodium and sulphur.

Schulz Flory plot of the product spectrum in Fig. 5 shows that the product spectrum follows the ideal Schulz Flory correlation with the exception of methane which is considerably lower than expected. The expected relation between methane and C_2 – C_4 selectivity as predicted by the Schulz Flory correlation does not apply for the Sasol Reference Na–S catalyst and this was found to be an effective methodology to evaluate the effect of non-metal promoters on C_2 – C_4 selectivity.

The positive deviation in the methane selectivity that is usually observed has been ascribed to unique methanation sites on the catalyst that cannot sustain chain growth and therefore produce methane in addition to that formed on the normal FT sites. It is thus tempting to explain the observed negative deviation in the methane selectivity in terms of a preferential blocking of these

methanation sites by sulphur. However, the data does not support such a theory. Even in the event that all hypothetical methanation sites are inactivated, the methane selectivity should still lie on the ideal Schulz Flory (S–F) line if the FT mechanism follows the ideal statistical distribution. The fact that the methane selectivity lies so far below the curve rather suggests that sulphur preferentially affects one of the reaction steps in the FT reaction mechanism. Noting that the termination of the C_1 intermediate to methane requires the addition of hydrogen, it is hypothesised that sulphur inhibits the hydrogenation steps more than it does the other reaction steps (such as carbon–carbon bond formation). This explains why, once the C_2 intermediate has been formed, chain growth continues with a more-or-less constant probability. It is also consistent with the observed high olefinicity of the product spectrum, as heavier intermediates would be more likely to terminate *via* a desorption step to the olefin than *via* hydrogenation to the paraffin. However it must also be noted that sodium plays a critical role in shifting the selectivity of the catalyst away from the ideal distribution. Fig. 6 shows the grouped selectivity of C_2 – C_4 against methane for catalysts with various levels of sulphur and sodium. In the case where the sodium is kept to a minimum (~ 1.5 mmol) and sulphur to a maximum (~ 4.5 mmol) one can see that the methane selectivity increases dramatically although the shift from ideal S–F distribution is still observed. As the amount of sodium is increased (max of ~ 26 mmol) the methane selectivity decreases, but with an associated decrease in C_2 – C_4 selectivity. Clearly, the combination of sodium and sulphur in the optimum concentration does yield the expected benefit, and high throughput screening assisted in identifying the ranges of each promoter required to achieve the desired selectivity. The results showed that there is a minimum ratio of 2:1 Na:S required to shift the selectivity. The optimum range for sodium is between 8 mmol and 26 mmol and for sulphur the optimum value is between 3 mmol and 4.5 mmol. The C_2 – C_4 olefin selectivity is

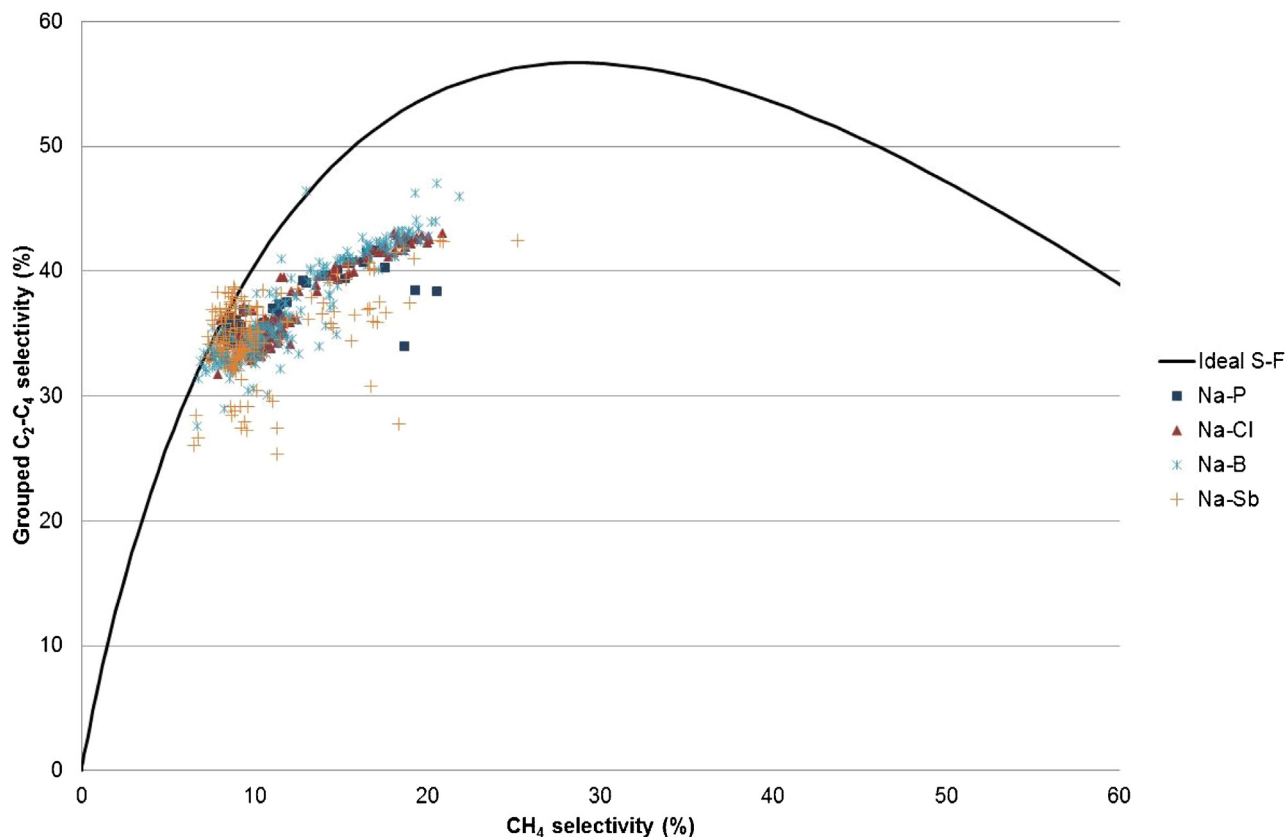


Fig. 7. Results obtained with Na in combination with P, Cl, B or Sb.

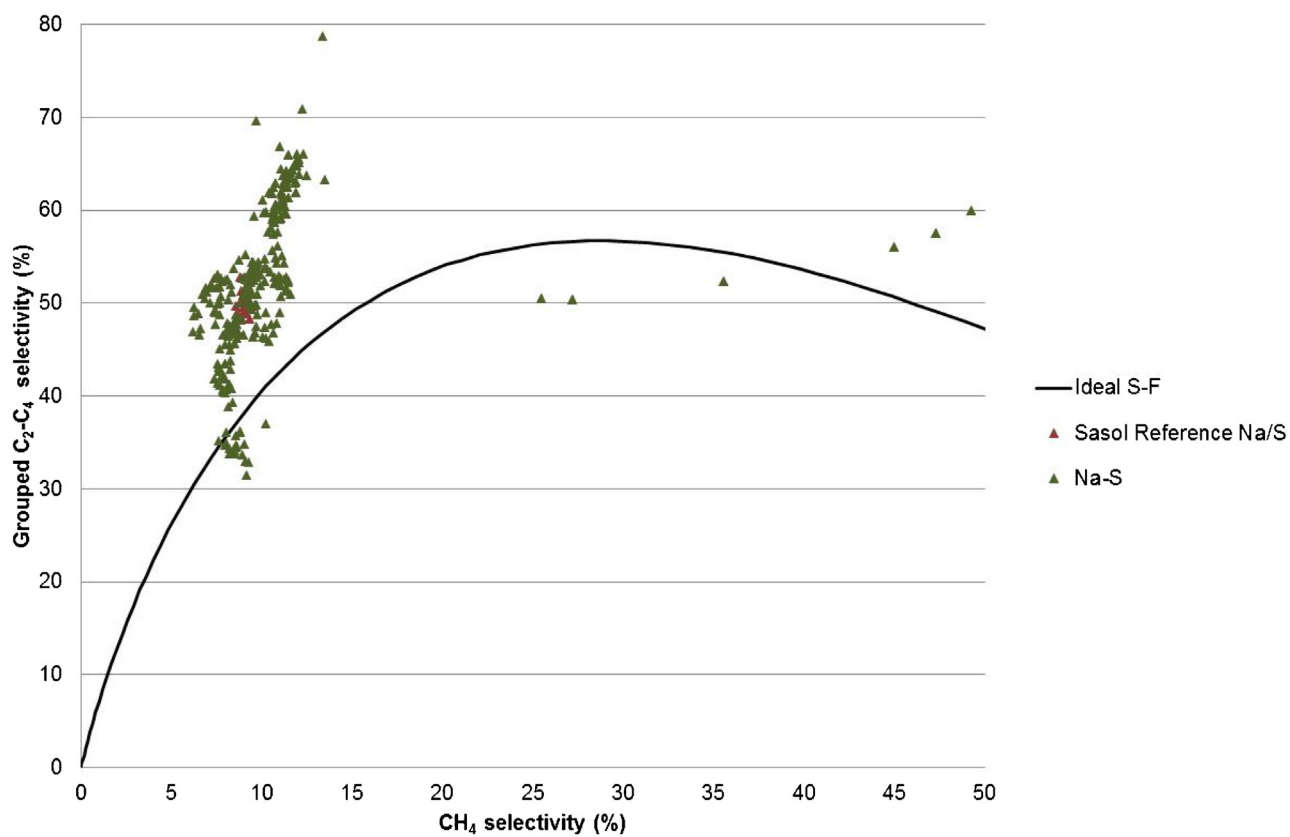


Fig. 8. Results obtained with Na in combination with S.

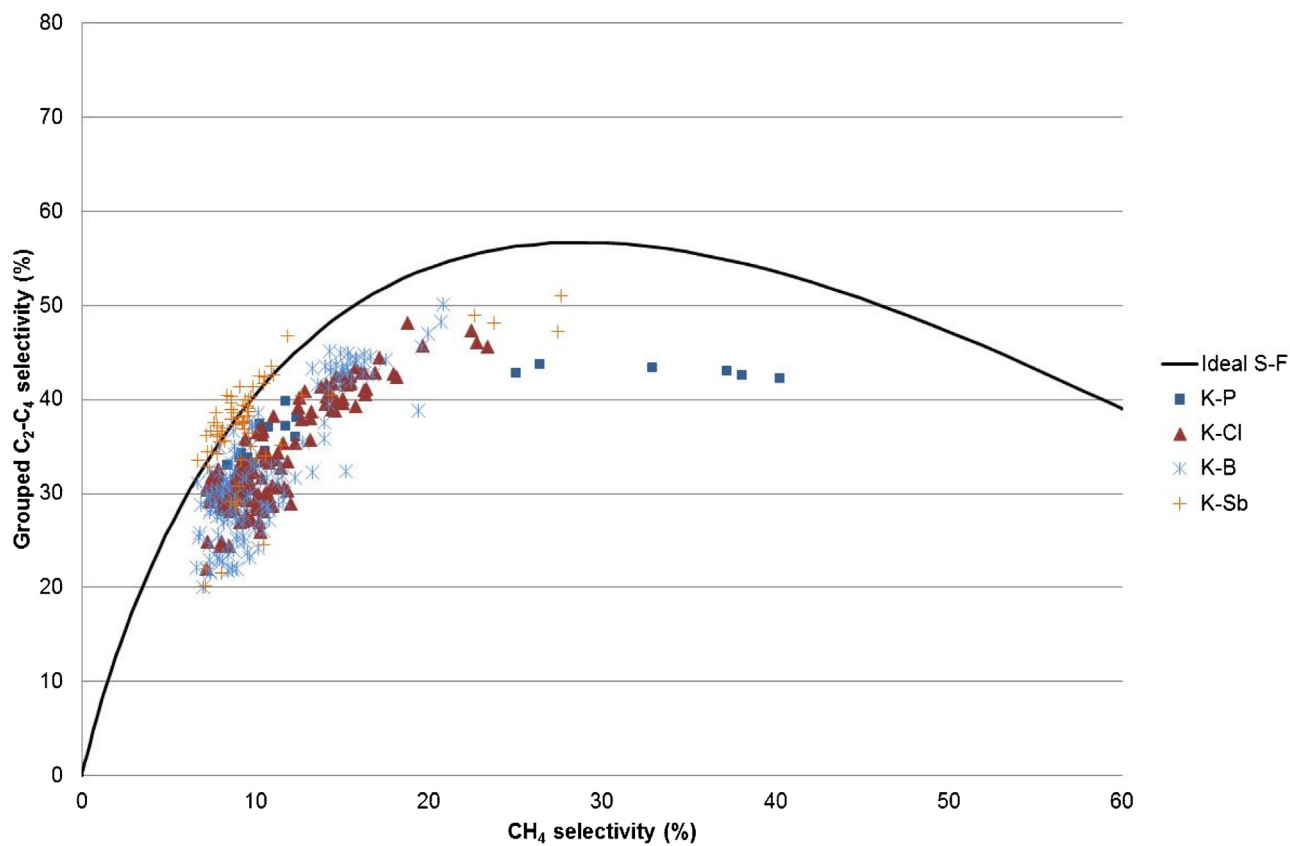


Fig. 9. Results obtained with K in combination with P, Cl, B or Sb.

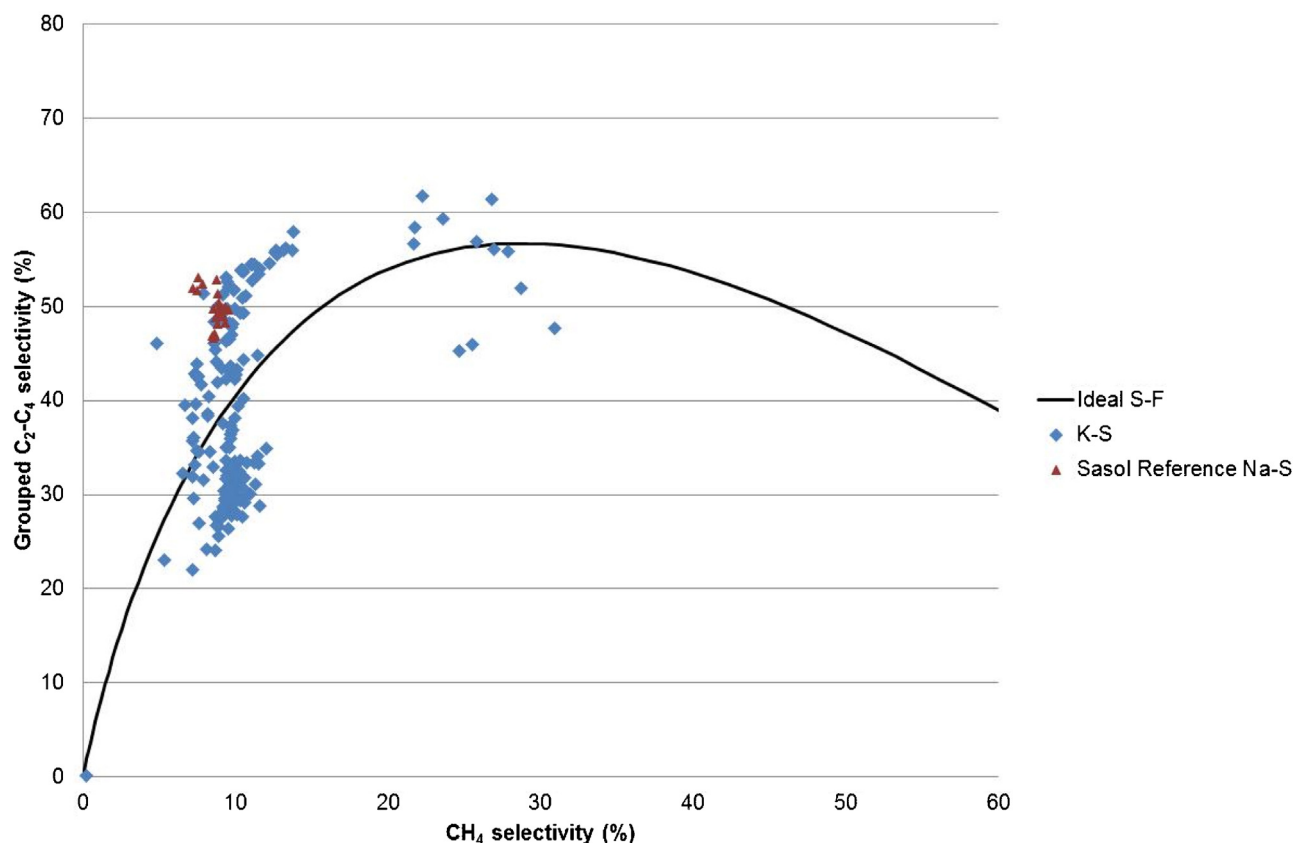


Fig. 10. Further results obtained with K in combination with S.

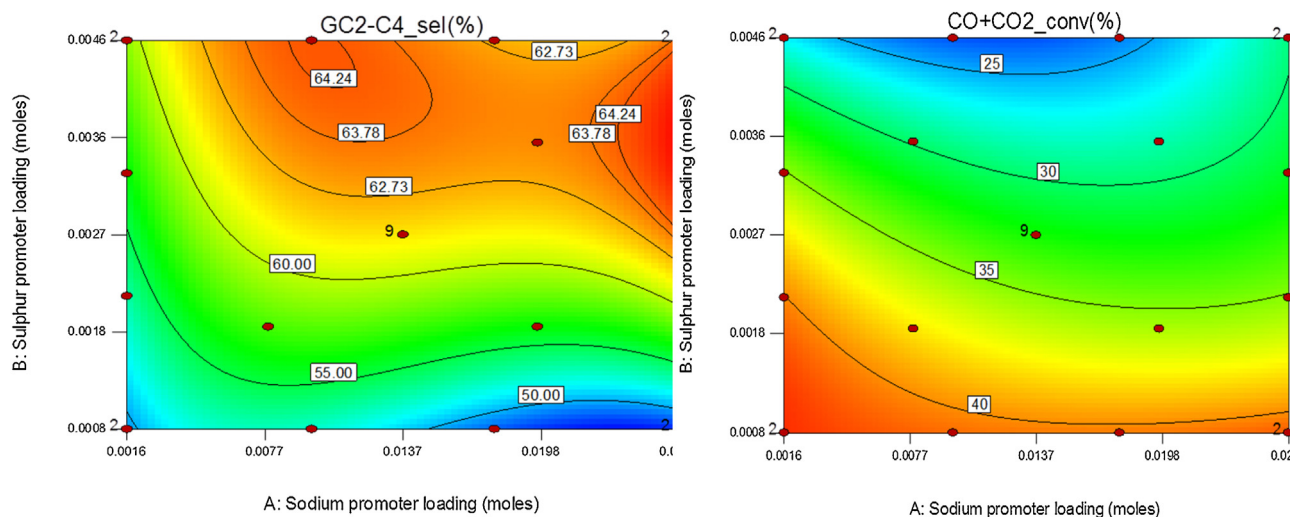


Fig. 11. Contour plots of predicted group C₂–C₄ selectivity (%) and CO conversion as a function of sodium and sulphur promoter loadings.

thus high for a wide range of sodium and sulphur concentrations and the selection of promoter levels would thus mainly depend on the required catalyst activity and stability.

3.2. Evaluation of alkali in combination with non-metal promoters

With the initial improved selectivity of C₂–C₄ and lower methane achieved with the Sasol Reference Na–S catalyst the question is whether the combination of sodium and sulphur is unique and whether other possible combinations of alkali and non-metal

promoters could potentially yield even more improved selectivities. Potassium was included in this evaluation due to the similar promotional effect compared to sodium. Based on initial work it was decided to select promoters from the group of elements collectively known as non-metals, semi-metals and halogens. Each of these promoters was combined with potassium and sodium to yield ten possible promoter combinations. Sodium (1.6–26 mmol) was tested in combination with each of the five non-metal promoters (phosphorous, chloride, boron and antimony) at a variety of levels (0.4–4.6 mmol). From Fig. 7, it is clear that phosphorous, chlorine and boron all yielded grouped C₂–C₄ selectivities on or below the

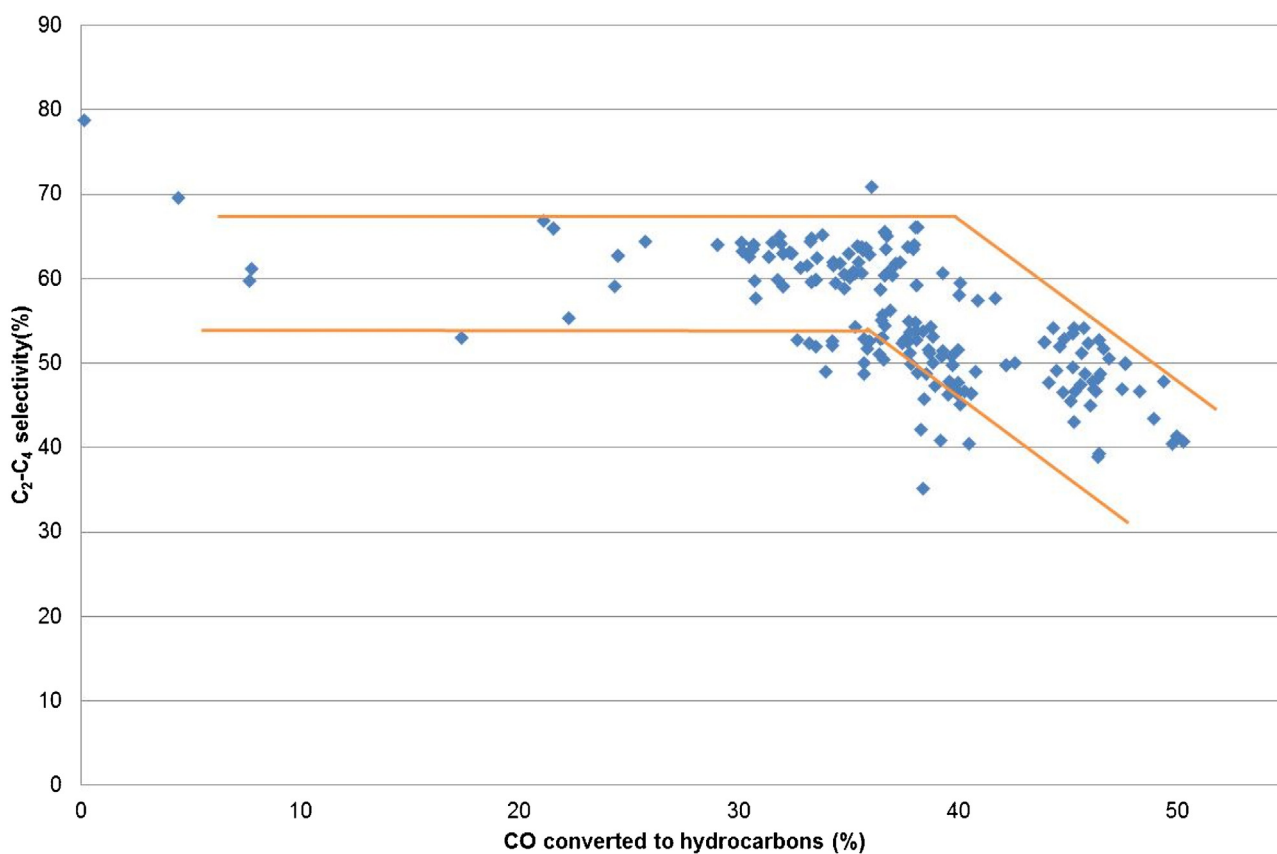


Fig. 12. The C₂-C₄ selectivities of various Na/S catalysts as a function of conversion.

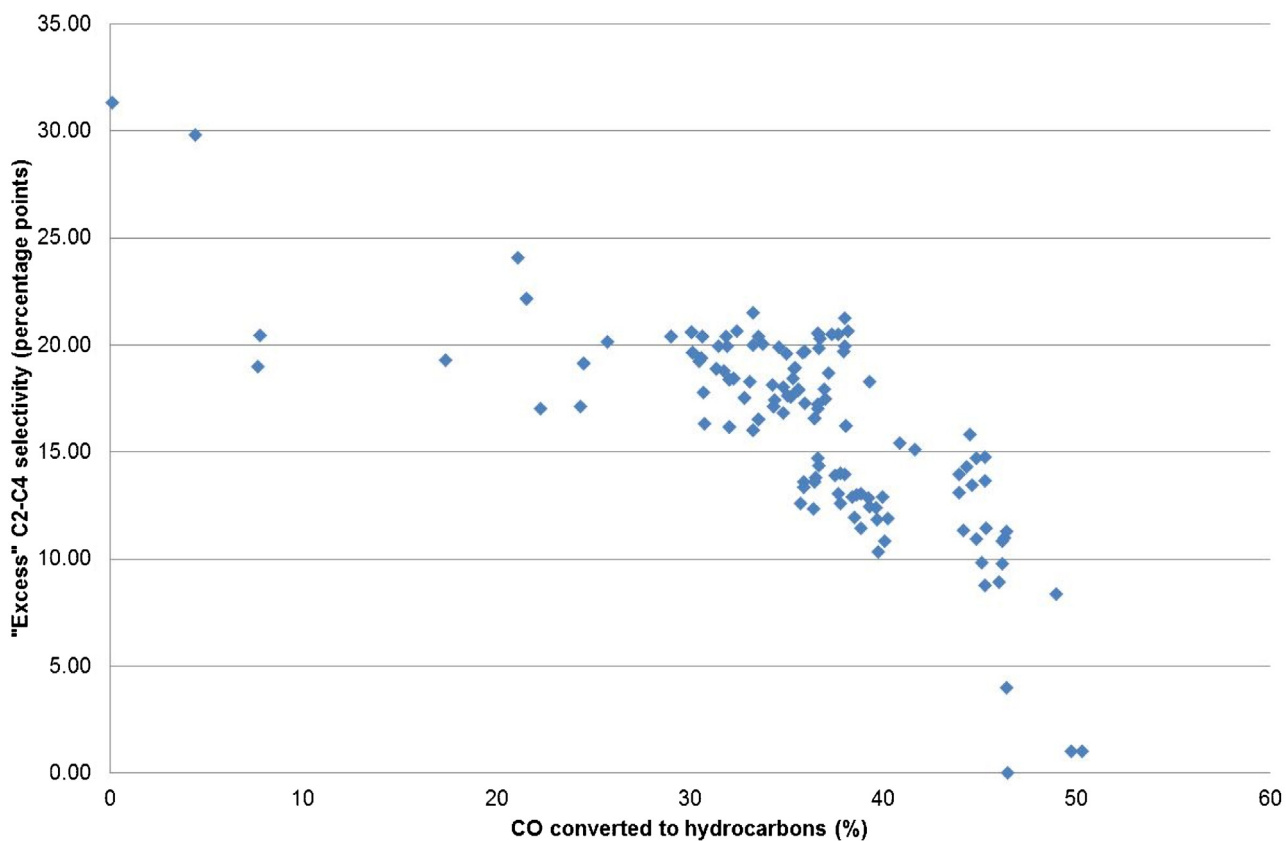


Fig. 13. The "excess" C₂-C₄ selectivities of various Na/S catalysts as a function of conversion.

curve predicted by the Schulz Flory correlation, indicating no significant improvement over traditional iron FTS catalysts. Although antimony showed small but notable improvements, because the grouped C_2 – C_4 selectivity for some of the catalyst formulations lies above the ideal relationship, the shift in selectivity was not significant enough to warrant further optimisation.

From Fig. 8 it is clear that none of the promoters evaluated yielded the dramatic improvement in C_2 – C_4 selectivity observed as in the case of sodium in combination of sulphur. The grouped C_2 – C_4 selectivity of some catalysts far exceeded the values expected from the ideal S–F (Schulz Flory) correlation. The data measured with the Sasol reference catalyst, which also contained Na and S, are included on this graph to show that significant selectivity improvements have been obtained with the high-throughput approach.

For the series of non-metal promoters evaluated in combination with potassium, the catalyst promoted with antimony and potassium yielded a slight shift in the C_2 – C_4 relationship with methane, but none of the other promoters showed any improvement in required selectivity (Fig. 9).

Again, out of the series of five non-metal promoters sulphur in combination with potassium, led to appreciable improvements in the grouped C_2 – C_4 selectivity over and above what can be achieved with traditional FTS (Fig. 10).

3.3. Evaluation of activity of Na/S catalysts with exceptional C_2 – C_4 selectivity

The Sasol reference Na–S catalyst containing sodium and sulphur yielded acceptable activity and CO conversion to hydrocarbon products. Unfortunately, the sodium–sulphur catalyst formulations evaluated through the high throughput study showed significantly lower CO conversion to hydrocarbons. The contour plot comparison in Fig. 11 indicates that maximum C_2 – C_4 selectivity is achieved for high promoter loadings however, for maximum CO conversion lower promoter loadings is required.

It is important to understand that since the gas inlet conditions were kept the same, the lower conversion is indicative of lower catalyst activity. If the C_2 – C_4 selectivity of catalysts is plotted against conversion it is observed that at high conversions (above 30% conversion) the C_2 – C_4 selectivity decreases dramatically (Fig. 12). The grouped C_2 – C_4 selectivity seems to decrease quite strongly with conversion when lower a CO conversion to hydrocarbons of about 30% is exceeded.

If the “excess” C_2 – C_4 selectivity (*i.e.* measured selectivity minus the value expected from ideal Schulz Flory correlation) is plotted as a function of conversion (Fig. 13) this relationship becomes clearer. The higher the conversion, the lower the “excess” grouped C_2 – C_4 selectivity, the more the product distribution comes in line with normal FT.

The observed relationship between the extent to which the new catalysts deviate from the ideal Schulz Flory correlation and the conversion measured could be ascribed to process conditions. Usually, the higher the conversion, the higher the exit (and average) H_2 /CO ratio and the more hydrogenating the environment becomes. However, the observed relationship can also be a result of the intrinsic catalyst properties. If the hypothesis is correct that the presence of sulphur in the catalyst selectively inhibits certain hydrogenation reaction steps, then per definition a catalyst with a good selectivity will have a low activity, since it is generally believed that the rate determining step of the FT reaction is a hydrogenation step. In isolation this would mean that targets to

increase the catalyst activity should be avoided, as this will destroy the selectivity benefits to the point where it becomes a traditional FT catalyst. However, evaluation of the impact of particle size on activity and selectivity cannot be ignored. A study on the impact of particle size on supported iron FTS catalysts showed that the size of the iron carbide particles had a significant impact on selectivity and activity [12]. The study focused on supported iron FTS catalysts including Na and S promoted systems. In this study it was found that an increase in promoter concentrations increased activity without negatively impacting selectivity. The evaluation of particle size on unsupported iron FTS catalysts is rather difficult due to the interaction of the surface iron with the bulk iron phases, however, further investigation into the impact of precipitation conditions on catalyst activity and selectivity is warranted in order to improve the conversion of the catalyst systems described above.

4. Conclusions

This work highlighted the application of high throughput experimentation to the accurate synthesis and screening of approximately 600 catalysts over a period of 8 months. The optimum catalyst formulation identified through this study showed a 25% improvement in C_2 – C_4 selectivity compared to Sasol's best in-house prepared precipitated iron catalyst. The methodology followed in this study, namely to plot the grouped C_2 – C_4 selectivity as a function of the methane selectivity together with the ideal Schulz Flory correlation, is a simple but quite effective evaluation criteria for the novel Na–S promoted FTS catalysts. If a significant “excess” amount of C_2 – C_4 components is produced the result will be a high overall selectivity to light chemicals (olefins and alcohols). However, it was shown in this work that the propensity of this novel group of catalysts to yield a product slate that deviates from the Schulz Flory correlation is significantly reduced at high catalyst activity and CO conversion levels. From the results it is clear that sulphur in combination with alkali (sodium or potassium) leads to a selectivity shift towards C_2 – C_4 hydrocarbon fraction without an increase in methane selectivity.

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References

- [1] H.M. Torres, K.P. de Jong, *ACS Catal.* 3 (2013) 2130–2149.
- [2] R.L. Espinoza, T. Shingles, D.J. Duvenhage, P.L. Langenhoven, U.S. Patent 6,653,357, Sasol Technology (2003).
- [3] B. Wu, L. Bai, H. Xiang, Y. Li, Z. Zhang, B. Zong, *Fuel* 83 (2004) 205–212.
- [4] Z. Tao, Y. Yang, C. Zhang, T. Li, M. Ding, H. Xiang, Y. Li, *J. Nat. Gas Chem.* 16 (2007) 278–285.
- [5] C.B. Murchison, D.A. Murrick, US Patent 4,151,190, The Dow Chemical Company (1979).
- [6] C.B. Murchison, US Patent 4,539,334, The Dow Chemical Company (1985).
- [7] W.L. Van Dijk, J.W. Niemantsverdriet, A.M. van der Kraan, H.S. van der Baan, *Appl. Catal.* 2 (1982) 273–288.
- [8] T.C. Bromfield, N.J. Coville, *Appl. Catal.* A 186 (1999) 297.
- [9] R. Crous, T.C. Bromfield, S. Booyens, International Publication No. WO2010/066386, 2010, Sasol Wax GmbH, Sasol Technology (Pty) Ltd. (2010).
- [10] H.M. Torres Galvis, A.C.J. Koeken, J.H. Bitter, T. Davidian, M. Ruitenbeek, I. Dugulan, K.P. de Jong, *J. Catal.* 303 (2013) 22–30.
- [11] Design-Expert®, Stat-East, Inc., 2021 East Hennepin Ave., Suite 480, Minneapolis, MN 55413 (2015).
- [12] H.M. Torres Galvis, J.H. Bitter, T. Davidian, M. Ruitenbeek, A.I. Dugulan, K.P. de Jong, *J. Am. Chem. Soc.* 134 (2012) 16207–16214.